

House Price Prediction using Linear Regression

Level 2 Task 1

Oasis Infobyte

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Track: Data Analytics

Introduction

House price prediction is a common regression problem in data analytics that helps estimate the market value of residential properties based on their characteristics. Accurate prediction of house prices assists buyers, sellers, and real estate professionals in making informed decisions.

The objective of this project is to build a Linear Regression model capable of predicting house prices using multiple property features. The project also compares the performance of Linear Regression with Ridge and Lasso Regression to evaluate the effect of regularization.

Objectives

- Perform Exploratory Data Analysis.
 - Handle categorical variables through One-Hot Encoding.
 - Build a Linear Regression model.
 - Evaluate the model using regression metrics.
 - Interpret the importance of different features.
 - Compare the performance with Ridge and Lasso Regression.
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Tools and Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Jupyter Notebook
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Methodology

The dataset was first explored to understand its structure, detect missing values, and generate descriptive statistics. Histograms and boxplots were used to study the distribution of numerical features, while a correlation heatmap was generated to identify relationships between variables.

Categorical variables were converted into numerical form using One-Hot Encoding. The data was then divided into training and testing sets using an 80:20 ratio. A Linear Regression model was trained using the training data, and predictions were generated for the testing dataset.

The model was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 Score. Additional visualizations such as the Actual vs Predicted plot and Residual plot were used to assess prediction accuracy and model assumptions. Finally, coefficient analysis was performed to understand the contribution of each feature, and Ridge and Lasso Regression models were trained for comparison.

Results

The Linear Regression model achieved:

- **MSE:** 1,754,318,687,330.71
- **RMSE:** 1,324,506.96
- **R^2 Score:** 0.6529

The Actual vs Predicted plot showed that most predictions were close to the ideal diagonal line, indicating good predictive performance. The Residual plot displayed randomly distributed residuals around zero, suggesting that the model satisfied the assumptions of Linear Regression reasonably well.

Coefficient analysis revealed that the number of bathrooms, air conditioning, preferred area, and the number of stories were among the strongest positive contributors to house prices, while unfurnished properties negatively influenced the predicted values.

The comparison between Linear Regression, Ridge Regression, and Lasso Regression showed nearly identical R^2 scores, indicating that regularization did not significantly improve performance for this dataset.

Conclusion

This project successfully demonstrated the complete workflow of a machine learning regression problem, from data exploration and preprocessing to model development and interpretation. The Linear Regression model explained approximately **65.3%** of the variation in house prices and produced reasonably accurate predictions.

Although Ridge and Lasso Regression were also evaluated, they achieved performance similar to the baseline Linear Regression model. Overall, the project highlights the importance of proper data preprocessing, feature analysis, and model evaluation in building effective predictive models for real-world housing data.